Zomato Delivery Time Estimation

A 10-hour ML sprint · DataVerse 2026 · BMSCE

Team [NAME] · P1 [n] P2 [n] P3 [n]
github.com/TheClazer/Zomato-delivery-estimation
EDA submission — 14:00 IST

What we're predicting

- Goal: predict Time_taken (min) for each delivery — given order, rider, restaurant, and conditions.
- Metric: Mean Absolute Error (MAE). Lower is better.
- Stakes: accurate ETAs drive customer trust and let Zomato balance courier load in real time.
- Baseline target: ≤4 min MAE; we are at 3.05 ± 0.02 min.

The data at a glance

- 45,153 rows × 21 columns (after coord + target filter)
- Target range: 10–54 min · median 26 · mean 26.3
- Top missing: Delivery_person_Ratings 3.7% · Delivery_person_Age 3.6% · Time_Orderd 3.2%

fig01 — Target distribution

How we cleaned the noise

- Stripped 'conditions ' prefix from weather, normalized 'NaN' strings → real NaN.
- Coerced numerics; parsed Order_Date with dayfirst=True.
- Capped impossible rider ratings (>5) → NaN.
- Dropped rows with zero-island coordinates (lat/lon both ≈0).
- Engineered haversine distance with >30 km physical cap (NaN, no row loss).

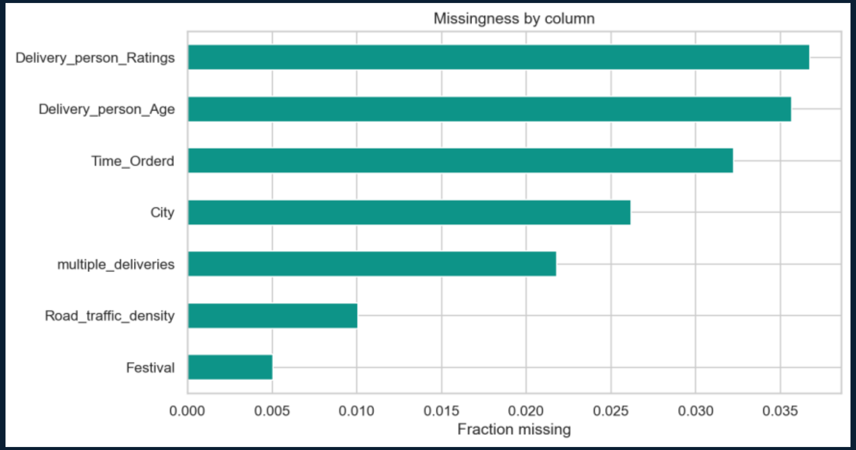


fig02 — Missingness

Distance is a clean primary signal

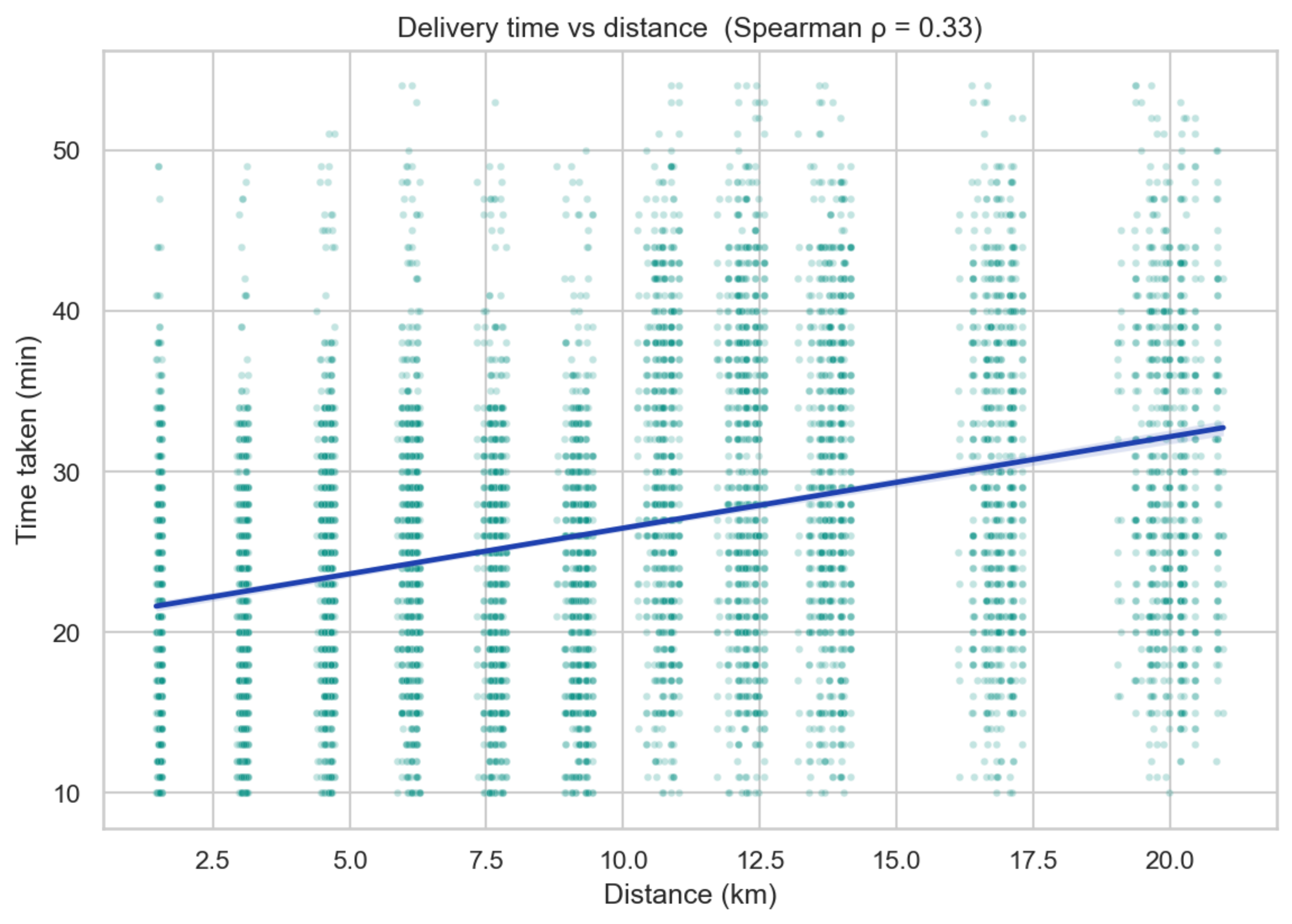
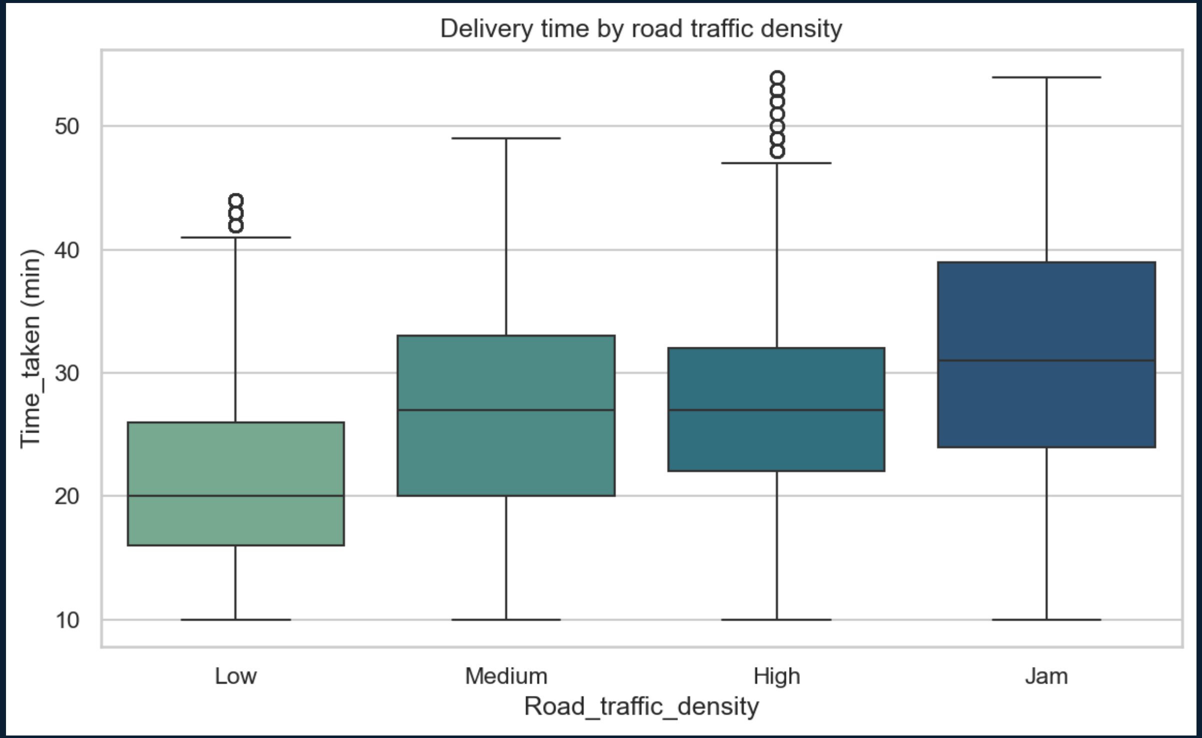
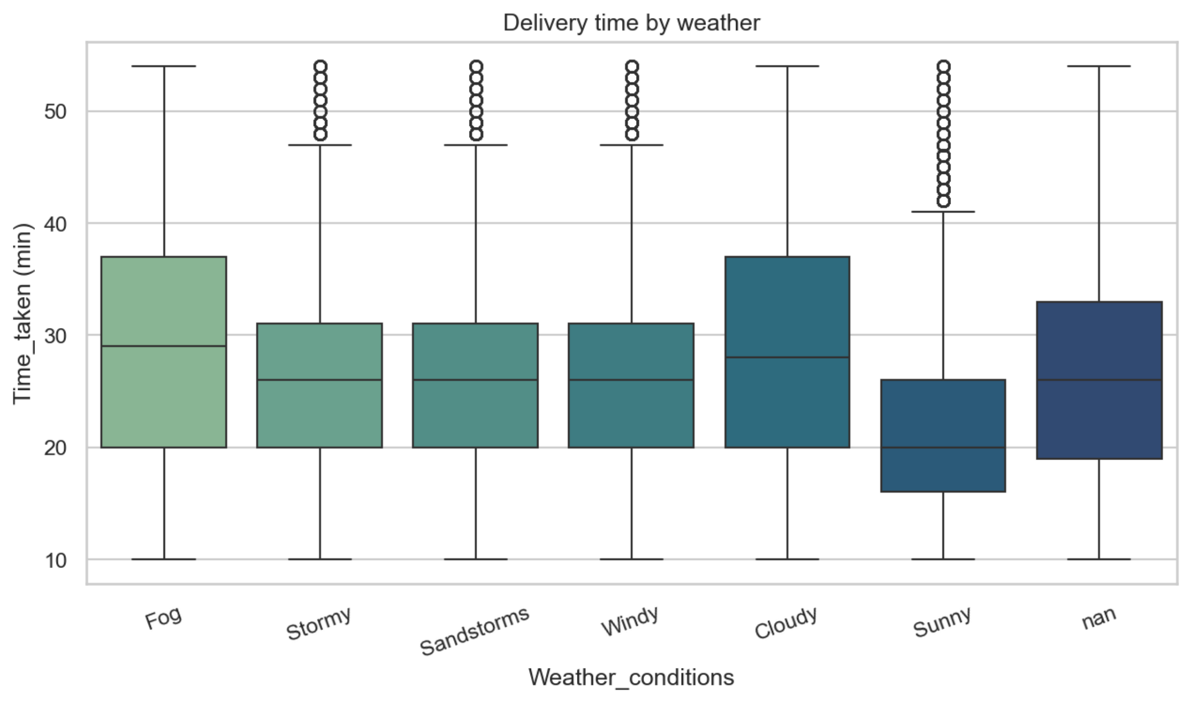


fig05 — distance vs time, regression line

- Spearman ρ = 0.33 between haversine distance and delivery time.
- Slope is positive and stable across folds.
- BUT — SHAP later shows distance is dominated by traffic; a finding the raw scatter hides.

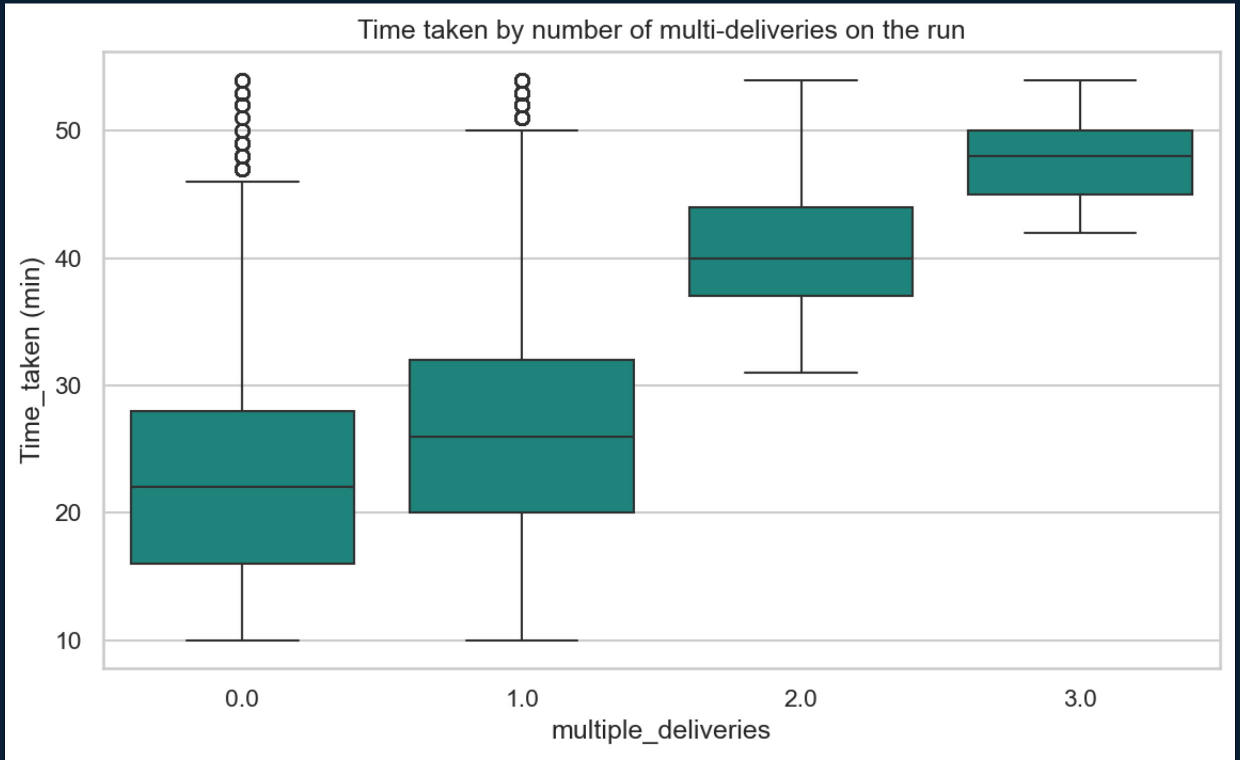
Traffic and weather compound the delay





- Traffic: groups differ — ANOVA F=3390.2, p=0.00e+00. Median grows Low → Jam.
- Weather: bad (Stormy/Fog/Sandstorm) > Sunny by ≈ 6.0 min median (p=0.00e+00).

Three smaller but real signals



- Multi-deliveries ↔ time: Spearman ρ = 0.34; each extra drop adds minutes.
- Festival days are ~19.5 min slower on average (Welch t, p=0.00e+00).
- Rider age has weak marginal effect once distance + rating are controlled.

The signal the raw data hides

- SHAP on the tuned LGBM ranks Road_traffic_density above distance.
- Distance is a strong univariate predictor (ρ≈0.32), but conditional on traffic the marginal effect shrinks.
- Operationally: an ETA model that ignores live traffic over-promises in conditions and under-promises in Low.

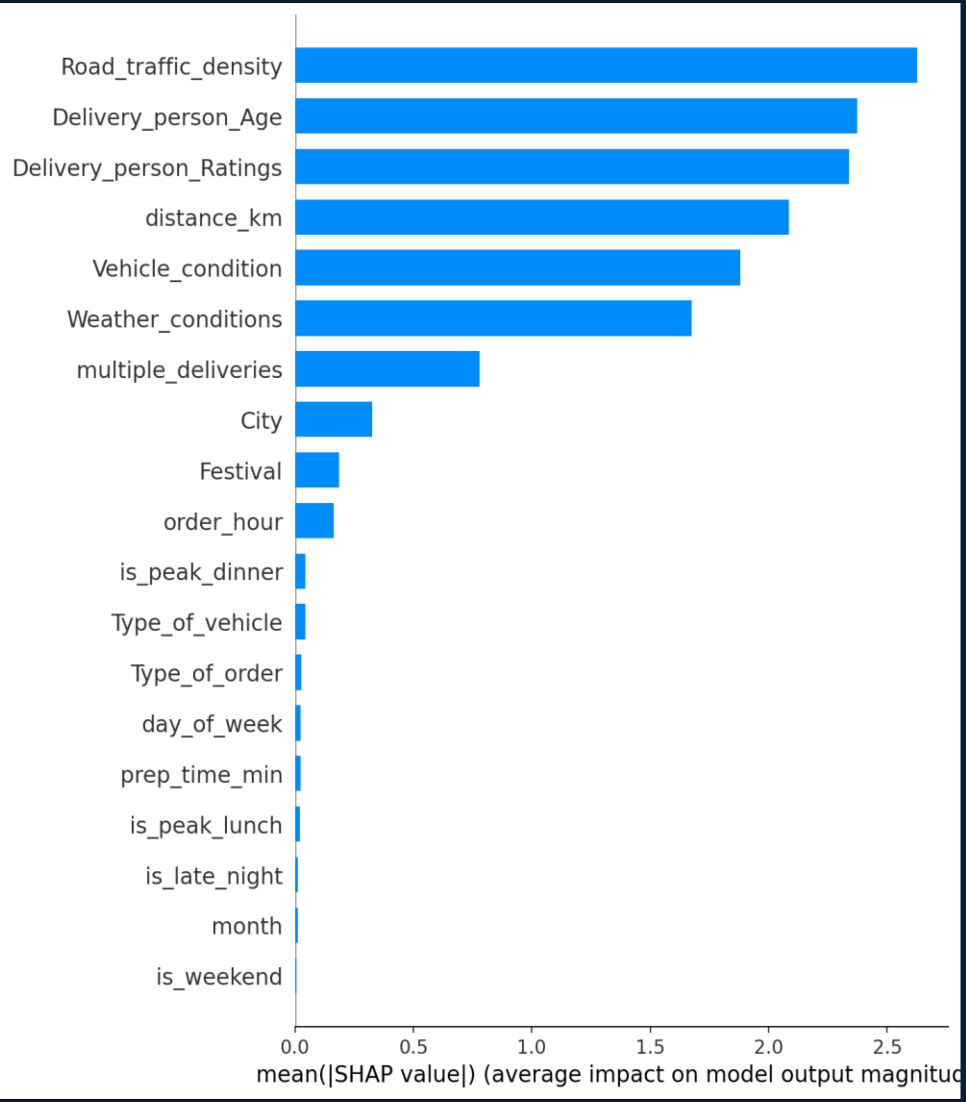


fig11 — SHAP feature importance on tuned LGBM v1

Features we built

- distance_km — haversine of pickup ↔ drop (capped at 30 km)
- prep_time_min — picked − ordered (midnight-safe)
- order_hour, is_peak_lunch, is_peak_dinner, is_late_night
- is_weekend, day_of_week, month
- 13 numeric + 6 categorical passed natively to LightGBM (no one-hot).

Hypothesis	Statistic	p	Verdict
H1: distance ↔ time (Spearma	ρ=0.321	p=0.00e+00	Weak
H2: traffic groups (ANOVA)	F=3390.2	p=0.00e+00	Different
H3: bad weather > sunny (M-W	U=109573900	p=0.00e+00	Yes
H4: festival > non-festival (Welc	t=138.32	p=0.00e+00	Yes
H5: multi-deliveries ↔ time (Sp	ρ=0.339	p=0.00e+00	Yes

fig09 — 5 hypothesis tests, all significant at p<0.05

What we'd do next

- Real-time traffic API as a feature (we already know it's the strongest signal).
- Restaurant prep-time history: rolling mean per restaurant_id.
- Quantile regression for ETA intervals, not just point predictions.
- Stack the existing blend: LGBM (3.05) + CatBoost (3.09) → blend 3.04.